Topic 13 — Archaeology

Covers syllabus (click to expand)

ASI · Stratigraphy · Carbon Dating · Excavations · Archaeological Sites · Archaeological Sites of Uttar Pradesh Taught so the method comes before the site: who runs the dig → how layers and dates work → who dug what → site↔state matches → UP trio.

Sources baked in: ASI history, excavation reports, UPPCS Prelims 2018–2025 **Weight:** ★★★ — UP IVC trio, 2023 site-state, Wakankar, 2020 Harappan match **Last verified:** August 2026 **Current Affairs:** N/A — static methods. Sanauli 2018 is the recent dig fact.

13.1 ASI

Archaeological Survey of India | 1861 | Union body under Culture

- The Archaeological Survey of India was founded in **1861** under Viceroy **Canning**.
- The first Director-General was **Alexander Cunningham**, who used Chinese pilgrim routes to hunt Buddhist sites such as Sarnath, Sanchi, and Bharhut.
- **John Marshall** served as DG from **1902 to 1928** and made ASI a standing scientific department.
- Marshall **announced** the Indus civilization to the world in **1924**.
- **Mortimer Wheeler** served as DG from **1944 to 1948** and brought the **balk-and-grid** excavation method.
- Wheeler re-dug Harappa in **1946**; that is a later method dig, not the discovery of the civilisation.
- ASI headquarters is **24 Tilak Marg, New Delhi**, not Lucknow.
- Lucknow holds UP **State** Archaeology and the State Museum — a different office from ASI HQ.
- Monument protection sits under the **AMASR Act 1958**; antiquities export sits under the **1972** Act.
- The Survey's journal is *Indian Archaeology — A Review*.
- Field units are regional **Circles** (for example Agra, Lucknow, and Patna), while **ASI HQ** remains New Delhi.

Logic: Cunningham founded the Survey. Marshall announced IVC. Wheeler did not discover it.

13.2 Stratigraphy

Layers in the trench | relative dating | deeper is older

- A habitation mound is a stack of floors, dumps, and fills built over time.
- By the **law of superposition**, the **bottom** layer is **older** than the top if the stack is not flipped by later pits or dumps.

- A **balk** is the uncut wall Wheeler leaves so the trench section can still be **read**.
- A **vertical** cut shows all periods in one chronology; a **horizontal** scrape exposes one floor and one culture horizon.
- **Typology** (प्रकारिकी) (pottery shape, tool type) sits beside layers as another relative clock.
- Stratigraphy alone cannot give "2600 BCE"; that needs an absolute clock such as C-14.

> **Logic:** Deeper ≠ always older if a pit is cut down from a later floor. Read the section, not the depth number alone.

13.3 Carbon Dating

Absolute date | organic carbon only | Libby

- **Willard Libby** opened the radiocarbon path that led to the 1949 Nobel recognition, and the method has been in use since the late 1940s.
- The **half-life** of C-14 is about **5730 years**.
- The method works on **charcoal, wood, bone, seed, and shell** — once-living samples that took in carbon.
- It does **not** date a stone tool or a copper sword by itself; date the charcoal or organic matter found **with** the object.
- The useful range is roughly to about 50,000 years; after that leftover C-14 is too thin to read well.
- Fired pottery can use **thermoluminescence (TL)** instead — a different clock and a different sample type.
- Mature Harappan chronology near **about 2600–1900 BCE** rests on absolute clocks plus pottery and stratigraphy, not guesswork from bricks alone.

> **Logic:** C-14 = organic. Stone handaxe ≠ a C-14 sample.

13.4 Excavations

Who opened which trench | discovery ≠ later re-dig

- In **1863**, **Robert Bruce Foote** at **Pallavaram** first recognised Indian Paleolithic tools.
- In **1921**, **Daya Ram Sahni** opened the first big IVC dig at **Harappa**.
- In **1922**, **R.D. Banerjee** opened urban IVC at **Mohenjo-daro**.
- In **1924**, **Marshall** as DG made the world announcement.
- In **1946**, **Wheeler** applied grid stratigraphy at Harappa.
- From **1955 to 1962**, **S.R. Rao** excavated **Lothal** and its dockyard.
- In **1957–58**, **V.S. Wakankar** established **Bhimbetka** as rock-art prehistory.
- In the **1960s**, **B.B. Lal** excavated **Kalibangan**, with the ploughed field and fire altars.

- In the **1990s**, **R.S. Bisht** excavated **Dholavira**, with reservoirs and the signboard.
- In **2018**, ASI under **S.K. Manjul** excavated **Sanauli** in Baghpat, with copper chariots and coffins.
- A find is useful only **in situ**, still in its layer.
- Looting kills the date.
- After 1947 the two giant IVC cities are in Pakistan.
- Indian trenches then shift to Gujarat, Haryana, Rajasthan, Punjab, and **Uttar Pradesh**.
- Boat **models** come from **Mohenjo-daro and Lothal** — not Dholavira alone.

PYQ — UPPCS Prelims 2020, Q10

Which of the following Indian Archaeologists first visited 'Bhimbetka Caves' and discovered prehistoric significance of its rock paintings?

A. Madho Swaroop Vatsa B. H.D. Sankalia C. V.S. Wakankar D. V.N. Mishra

Show answer

Ans: C. Vatsa = Harappan trenches. Sankalia = Deccan prehistory (Langhnaj). Bhimbetka is in **Madhya Pradesh**.

PYQ — UPPCS Prelims 2022, Q68

From which archaeological site of the Indus Valley Civilization are figures or models of boats found?

A. Dholavira and Bhagatrav B. Harappa and Kot Diji C. Mohenjo-daro and Lothal D. Kalibangan and Ropar

Show answer

Ans: C.

> **Logic:** Wakankar ≠ Vatsa. Wheeler ≠ Sahni.

13.5 Archaeological Sites

Paper wants present state | two match-lists to freeze: prehistoric site → state

Site	State
Nevasa	**Maharashtra** (Pravara)
Isampur	**Karnataka** (Acheulian)
Didwana	**Rajasthan** (थार) (Thar fringe)
Gudiyam Cave	**Tamil Nadu**

PYQ — UPPCS Prelims 2023, Q27

Match List-I with List-II.

A. Nevasa B. Isampur C. Didwana D. Gudiyam Cave 1. Rajasthan 2. Tamil Nadu 3. Maharashtra 4. Karnataka

A. A-3, B-4, C-1, D-2 B. A-4, B-3, C-1, D-2 C. A-3, B-4, C-2, D-1 D. A-3, B-2, C-4, D-1

Show answer

Ans: A.

Harappan site → state

Site	State / UT
Balu	**Haryana**
Manda (मांडा)	**Jammu & Kashmir**
Padri	**Gujarat**
Hulas	**Uttar Pradesh**

PYQ — UPPCS Prelims 2020, Q12

Match List-I (Harappan Site) with List-II (UT/State of India).

A. Balu B. Manda C. Padri D. Hulas 1. Uttar Pradesh 2. Jammu & Kashmir 3. Haryana 4. Gujarat

A. 3 2 1 4 B. 2 3 4 1 C. 2 4 3 1 D. 3 2 4 1

Show answer

Ans: D — 3 2 4 1.

- **Bhimbetka** is in **Madhya Pradesh** as a UNESCO rock-shelter complex, not a Uttar Pradesh site.
- **Rakhigarhi** lies in **Hisar, Haryana**, and is the largest Indus mound — it is **not** in Uttar Pradesh.
- **Mehrgarh** is in Balochistan and marks pre-Harappan farming; it is not an Indian-state match option.

> **Logic:** Manda ≠ Mandi. Padri ≠ Hulas.

13.6 Archaeological Sites of Uttar Pradesh

IVC east fringe + doab mounds + Buddhist / Mauryan UP sites (learn cold)

Site	District / river	Fact
Alamgirpur	Meerut, **Hindon**	**Easternmost** IVC site in India
Hulas	Baghpat	UP Harappan; PYQs **2018, 2020, 2025**

Site	District / river	Fact
Mandi	Bijnor, **Ramganga** (रामगंगा)	UP Harappan (2021, 2025)
Often used as bait	Actual location	
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Manda	Jammu and Kashmir	
Rakhigarhi	Haryana (largest IVC mound in India)	
Kalibangan	Rajasthan	
Lothal	Gujarat	
Balu	Haryana	

Other UP trenches

- **Sanauli** in Baghpat, excavated in **2018**, yielded copper **chariots**, coffins, and antenna swords in a Late Harappan / OCP setting.
- It sits in the same district as Hulas, but the period label is different.
- **Koldihwa** and **Mahagara** on the Belan near Prayagraj are Neolithic sites with an early **rice** claim.
- **Hastinapur** in Meerut is a PGW and Kuru mound linked with Lal's flood-layer story.
- **Kaushambi** near Prayagraj is the Vatsa capital with a Mauryan–Gupta stack; the Allahabad pillar stone was moved from here.
- **Ahichchhatra** in Bareilly is the Panchala mound.
- **Sarnath** at Varanasi is the first-sermon site with the Ashokan lion capital and was one of Cunningham's hunting grounds.
- **Rajghat** preserves the old Kashi rampart.
- **Shravasti** at Sahet-Mahet is the Kosala and Jetavana card.
- **Atranjikhera** in Etah is a PGW type-site.

PYQ — UPPCS Prelims 2025, Q87

Which of the following archaeological sites related to the Indus Valley Civilization are situated in present-day Uttar Pradesh?

1. Mandi 2. Rakhigarhi 3. Hulas

A. 1 and 2 B. Only 3 C. 1 and 3 D. Only 1

Show answer

Ans: C. Rakhigarhi is Haryana.

PYQ — UPPCS Prelims 2023, Q28

The eastern boundary of the Harappan culture is indicated by which of the following?

A. Harappa B. Alamgirpur C. Rakhigarhi D. Manda

Show answer

Ans: B. Harappa is in Pakistan. Rakhigarhi is big, not east. Manda is north (J&K).

PYQ — UPPCS Prelims 2021, Q100

In which State of India is the Harappan Civilization site Mandi situated?

A. Gujarat B. Haryana C. Rajasthan D. Uttar Pradesh

Show answer

Ans: D. Bijnor.

PYQ — UPPCS Prelims 2018, Q88

Which of the following centres related to Indus Valley are situated in Uttar Pradesh?

I. Kalibanga II. Lothal III. Alamgirpur IV. Hulas

A. I, II, III, IV B. I, II C. II, III D. III, IV

Show answer

Ans: D. Kalibangan = Rajasthan. Lothal = Gujarat.

> **Logic:** Eastern IVC end in India = **Alamgirpur** (Meerut). Largest mound = **Rakhigarhi** (Haryana).

13.7 Archaeological Cultures — Pottery and Metal Horizons

Doab sequence memory | pottery labels are period clocks

- **OCP** (Ochre Coloured Pottery) marks the eastern doab Chalcolithic horizon and often pairs with **Copper Hoard** finds.
- **PGW** (Painted Grey Ware) is the classic **Later Vedic** north Indian ware; Hastinapur and Atranjikhera are type mounds.
- **NBPW** (Northern Black Polished Ware) rises with second urbanisation from about the **sixth century BCE** and remains thick in **Mauryan** cities.
- **Megalithic** burials with large stones belong mainly to the Iron-age **south**; they are not a Harappan city label.
- Safe doab memory order is **Late Harappan / OCP → PGW → NBPW**.

Culture	Marker	Usual date / link
OCP (Ochre Coloured Pottery)	eastern doab pottery	Chalcolithic overlap; pairs with **copper hoards**
PGW (Painted Grey Ware)	grey painted bowls	**Later Vedic** north; Hastinapur, Atranjikhera

Culture	Marker	Usual date / link
NBPW (Northern Black Polished Ware)	lustrous northern fine ware	~6th c. BCE rise; **Mauryan** cities
Megalithic	burials with big stones	Iron-age **south**; not Harappan

13.8 Major Excavation Sites — Signature Cards

One site = one sentence | state first, then signature

- **Rakhigarhi** in Haryana is the largest Indus mound in India and is **not** in Uttar Pradesh.
- **Dholavira** in Gujarat, excavated by **R.S. Bisht**, is remembered for reservoirs, the stone signboard, and three-part town planning.
- **Kalibangan** in Rajasthan, excavated by **B.B. Lal**, is remembered for the ploughed field and fire altars.
- **Banawali** in Haryana is a fortified Harappan town with a more radial street plan.
- **Surkotada** in Gujarat is a small Harappan fort whose horse-bone claim remains disputed.

Site	State	Excavator / fact	Signature find
Rakhigarhi	Haryana	recent ASI digs	largest IVC mound in India; **not UP**
Dholavira	Gujarat	R.S. Bisht	reservoirs, **signboard**, stone architecture
Kalibangan	Rajasthan	B.B. Lal	ploughed field, fire altars
Banawali	Haryana	—	fortified Harappan town
Surkotada	Gujarat	—	Harappan fort; horse-bone debate

13.9 Dating Methods — Beyond C-14

Relative clocks answer older/younger | absolute clocks answer calendar years

- **Stratigraphy** reads soil layers for relative order only.
- **Typology** matches pottery and tool shapes across sites as another relative clock.
- **Radiocarbon (C-14)** dates organic samples (charcoal, bone, seed) using a half-life of about **5730 years**.
- **Thermoluminescence (TL)** dates fired pottery or bricks by asking when the object was last heated.
- Use relative methods to order a trench; use C-14 or TL when the stem wants a calendar band.

Method	Samples	Fact
Stratigraphy	soil layers	relative order only
Typology	pottery shape	relative, cross-site

Method	Samples	Fact
Radiocarbon (C-14)	charcoal, bone, seed	organic only; ~5730 yr half-life
Thermoluminescence (TL)	fired pottery, bricks	when object was last heated

UP Focus

UP site facts already sit in the table above. Extra traps:

Trap	Fact
Bhimbetka in UP	**MP**
ASI HQ Lucknow	**New Delhi**
Sanauli = Mature Harappan dock	Late Harappan / OCP **chariots**, Baghpat